Lab3 competition ROCstars

Emery Berry, Ryan Morreale, Jacob Krol

2026-03-27

Instructions

- define prof.test
- run all

```
prof.test <- NULL
```

Libraries

```
library(e1071)
library(ggplot2)
```

```
Attaching package: 'ggplot2'

The following object is masked from 'package:e1071':

    element
```

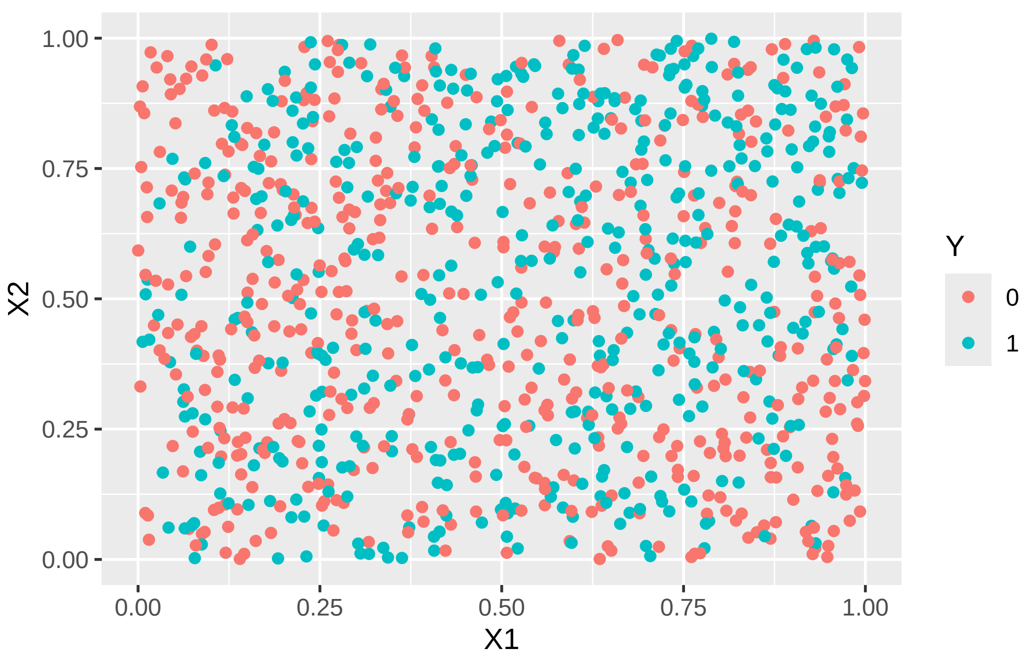
Train generation

```
generate_y <- function(x1,x2) { #two input parameters to generate the output y
  logit <- x1 -2*x2 -2*x1^2 + x2^2 + 3*x1*x2
  p <- exp(logit)/(1+exp(logit)) #apply the inverse logit function
  y <- rbinom(1,1,p) #y becomes a 0 (with prob 1-p) or a 1 with probability p
}
```

```r
# Generate a training dataset with 1000 points
set.seed(321)
n = 1000
X1_train <- runif(n,0,1)
X2_train <- runif(n,0,1)

#I'm going to use a for loop to generate 1000 y's
Y_train <- rep(0,n) #initializing my Y to be a vector of 0's
for (i in 1:n) {
  Y_train[i] <- generate_y(X1_train[i],X2_train[i])
}

sum(Y_train)
```

```
[1] 465
```

```r
training <- data.frame(X1=X1_train, X2=X2_train, Y=as.factor(Y_train))
ggplot(data=training, aes(x=X1, y=X2, color=Y)) +
  geom_point()
```



Fit team SVM

```r
svm_radial <- svm(Y ~ ., data=training, kernel="radial", gamma=0.1, cost=1)
summary(svm_radial)
```

```
Call:
svm(formula = Y ~ ., data = training, kernel = "radial", gamma = 0.1,
    cost = 1)


Parameters:
   SVM-Type:  C-classification
 SVM-Kernel:  radial
       cost:  1

Number of Support Vectors:  872

 ( 437 435 )


Number of Classes:  2

Levels:
 0 1
```

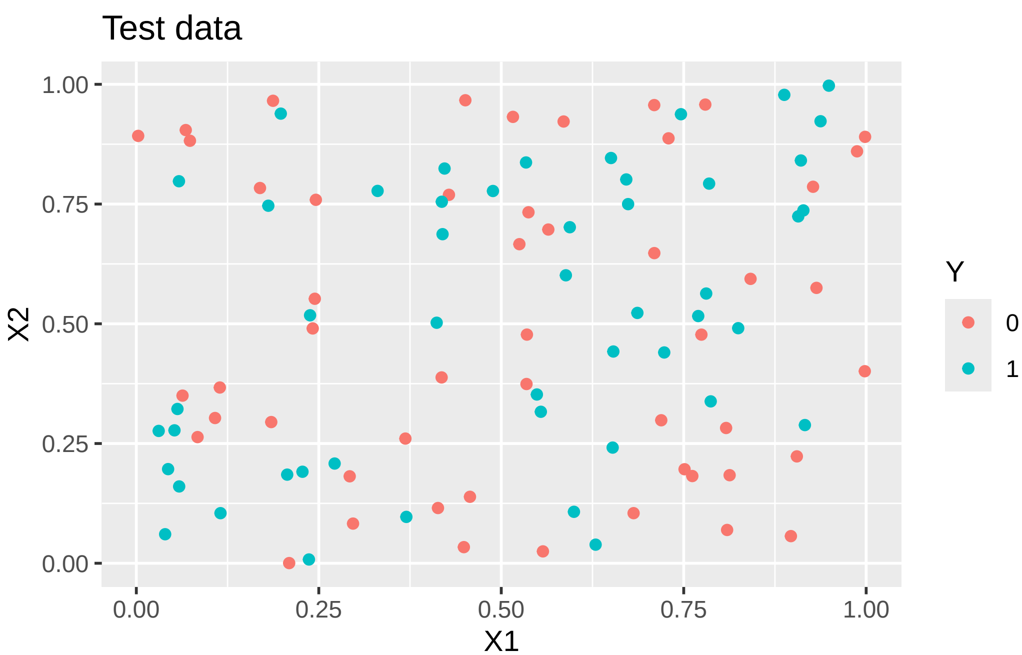
Test evaluation

Generate test if not provided (for testing purposes)

```r
if (is.null(prof.test)) {
  cat("No test data provided, generating test data\n")
  SEED <- 323

  set.seed(SEED)

  n = 100
  X1_test <- runif(n,0,1)
  X2_test <- runif(n,0,1)
```

```r
  Y_test <- rep(0,n)
  for (i in 1:n) {
    Y_test[i] <- generate_y(X1_test[i],X2_test[i])
  }

  prof.test <- data.frame(X1=X1_test, X2=X2_test, Y=as.factor(Y_test))

  cat("# of 1s", sum(Y_test), "\n")
  cat("# of 0s", sum(1-Y_test), "\n")
  ggplot(data=prof.test, aes(x=X1, y=X2, color=Y)) +
    geom_point() +
    ggtitle("Test data")
} else {
  cat("Using provided test data\n")
  prof.test
}
```

```
No test data provided, generating test data
# of 1s 49
# of 0s 51
```



Evaluate team SVM

```r
# set the column names to match the training data
pred <- predict(svm_radial, newdata=data.frame(X1=prof.test[,1], X2=prof.test[,2], Y=prof.tes
sens <- sum(pred == 1 & prof.test[,3] == 1) / sum(prof.test[,3] == 1)
spec <- sum(pred == 0 & prof.test[,3] == 0) / sum(prof.test[,3] == 0)
misclass_error <- mean(pred != prof.test[,3])
cat("Sensitivity:", sens, "\n")
```

```
Sensitivity: 0.3469388
```

```r
cat("Specificity:", spec, "\n")
```

```
Specificity: 0.7058824
```

```r
cat("Misclassification error:", misclass_error, "\n")
```

```
Misclassification error: 0.47
```